Solubility chart

A **solubility chart** is a chart describing whether the ionic compounds formed from different combinations of cations and anions dissolve in or precipitate from solution.

Chart

The following chart shows the solubility of various ionic compounds in water at 1 atm pressure and room temperature (approx. 25 °C, 298.15 K). "Soluble" means the ionic compound doesn't precipitate, while "slightly soluble" and "insoluble" mean that a solid will precipitate; "slightly soluble" compounds like calcium sulfate may require heat to precipitate. For compounds with multiple hydrates, the solubility of the most soluble hydrate is shown.

Some compounds, such as nickel oxalate, will not precipitate immediately even though they are insoluble, requiring a few minutes to precipitate out.[1]

Key

S	highly soluble or miscible	≥20 g/L
sS	slightly soluble	0.1~20 g/L
I	relatively insoluble	<0.1 g/L
R	reacts with or in water	—
?	unavailable	—

Ions names and symbols	Halogens					Chalcogens				Pnictogens		
	Fluoride F^-	Chloride Cl^-	Bromide Br^-	Iodide I^-	Perchlorate ClO_4^-	Oxide O^{2-}	Hydroxide OH^-	Sulfide S^{2-}	Sulfate SO_4^{2-}	Nitrate NO_3^-[a]	Phosphate PO_4^{3-}	Carbonate CO_3^{2-}[a]
Hydrogen H^+	S	S	S	S	S	S	S	sS	S	S	S	S
Ammonium NH_4^+[a]	S	S	S	S	S	S[b]	S	R	S	S	S	S
Lithium Li^+	sS	S	S	S	S	R	S	R	S	S	sS	sS
Sodium Na^+	S	S	S	S	S	R	S	R	S	S	S	S
Potassium K^+	S	S	S	S	sS	R	S	R	S	S	S	S
Rubidium Rb^+	S	S	S	S	sS	R	S	R	S	S	S	S
Caesium Cs^+	S	S	S	S	sS	R	S	R	S	S	S	S
Beryllium Be^{2+}	S	S	S	R	S[4]	I	I	R	S	S	I	sS
Magnesium Mg^{2+}	sS	S	S	S	S	R	I	R	S	S	I	sS
Calcium Ca^{2+}	I	S	S	S	S	R	sS	R	sS	S	I	I
Strontium Sr^{2+}	sS	S	S	S	S	R	sS	R	sS	S	sS	I
Barium Ba^{2+}	sS	S	S	S	S	R	S	R	I	S	I[9]	sS
Aluminium Al^{3+}	sS	S	S	S[c]	S[10]	I	I	R	S	S	I	R
Gallium Ga^{3+}	I	S	S	R	S[10]	I	I	R	sS	S	I	R
Manganese(II) Mn^{2+}	sS	S	S	S	S[13]	I	I	I	S	S	I	I
Iron(II) Fe^{2+}	sS	S	S	S	S	I	I	I	S	S	I	I
Cobalt(II) Co^{2+}	sS	S	S	S	S[15]	I	I	I	S	S	I	I
Nickel(II) Ni^{2+}	S	S	S	S	S	I	I	I	S	S	I	I
Copper(II) Cu^{2+}	sS	S	S	?	S	I	I	I	S	S	I	I[d]
Zinc Zn^{2+}	sS	S	S	S	S[16]	I	I	I	S	S	I	I
Cadmium Cd^{2+}	S	S	S	S	S[18]	I	I	I	S	S	I	I
Mercury(II) Hg^{2+}	R	S	sS	I	S[19]	I	I	I	R	S	I	I
Vanadium(III) V^{3+}	I	S	S	S	S[21]	I	I	I	sS	S	I	?
Chromium(III) Cr^{3+}	sS	S	S	S	S	I	I	I	S	S	I	I
Iron(III) Fe^{3+}	S[e]	S	S	R	S[22]	I	I	I	S	S	sS	R[23]
Gold(III) Au^{3+}	R	S	sS	?	?	I	I	I	?	?	I	I
Tin(II) Sn^{2+}	S	S	S	S	S[26]	I	I	I	S	?	I	I
Lead(II) Pb^{2+}	sS	sS	sS	sS	S	I	sS	I	I	S	I	I
Silver Ag^+	S	I	I	I	S	I	I	I	sS	S	I	I
Mercury(I) Hg_2^{2+}	R	I	I	I	S[28]	I	?	?	sS	S[g]	?	I
	Fluoride F^-	Chloride Cl^-	Bromide Br^-	Iodide I^-	Perchlorate ClO_4^-	Oxide O^{2-}	Hydroxide OH^-	Sulfide S^{2-}	Sulfate SO_4^{2-}	Nitrate NO_3^-[a]	Phosphate PO_4^{3-}	Carbonate CO_3^{2-}[a]

See also

- Solubility rules

Notes

a. Compounds that include ammonium (NH_4^+), chlorate (ClO_3^-), or nitrate (NO_3^-) are soluble without exceptions. Compounds that include carbonate (CO_3^{2-}) are insoluble, unless the compound includes group 1 elements or ammonium.[2]

b. "Ammonium oxide" does not exist. However, its theoretical molecular formula $(NH_4^+)_2O^{2-}$ represents that of aqueous ammonia.

c. Partial electrolysis.

d. The commonly encountered basic copper carbonate ($Cu_2CO_3(OH)_2$) is insoluble in water. True copper(II) carbonate ($CuCO_3$) is rare and reacts with water to form basic copper carbonate.

e. Anhydrous FeF_3 is slightly soluble in water; $FeF_3 \cdot 3H_2O$ is much more soluble in water.

f. The commonly encountered basic iron(III) acetate ($[Fe_3O(OAc)_6(H_2O)_3]OAc$) is insoluble in water. True iron(III) acetate ($Fe(OAc)_3$) is rare and is soluble in water.

g. Slowly decomposes in water.

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